

A Counterexample for Detecting Almost Δ -Points on the Unit Sphere

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Status

This packet records a candidate counterexample, likely valid. It answers a question raised by Choi and Jung in arXiv:2307.10647, page 9: whether the almost- Δ analogue of their Proposition 2.12 holds in general.

Source Passage

The source paper defines almost Δ -points by requiring that for every $\varepsilon > 0$ there is $x \in B_X$ such that

$$x \in \overline{\text{co}} \Delta_\varepsilon(x), \quad \Delta_\varepsilon(x) = \{y \in B_X : \|y - x\| \geq 2 - \varepsilon\}.$$

After proving that almost Daugavet points are detected by $\sup_{x \in S_X} dc(x) = 2$, the authors write that it is unclear whether the analogous result for almost Δ -points holds in general.

X admits almost Daugavet points (resp. almost Δ -points) if for every $\varepsilon > 0$, there exists $x \in B_X$ such that $B_X \subseteq \overline{\text{co}} \Delta_\varepsilon(x)$ (resp. $x \in \overline{\text{co}} \Delta_\varepsilon(x)$). From Proposition 2.3, we have the following characterization, which shows that the behavior of the Daugavet constant of points in the unit sphere S_X completely determines whether X admits almost Daugavet points or not.

Proposition 2.12. *Let X be a Banach space. Then X admits almost Daugavet points if and only if $\sup_{x \in S_X} \text{dc}(x) = 2$.*

Proof. The proof of the “if” part is clear. Assume that X admits almost Daugavet points. Let $\varepsilon > 0$ and $x_0 \in B_X$ be so that $B_X \subseteq \overline{\text{co}} \Delta_{\varepsilon/2}(x_0)$. By Proposition 2.3, it is clear that $\text{dc}(x_0) \geq 2 - \frac{\varepsilon}{2}$. Note that $\|x_0\| \geq 1 - \frac{\varepsilon}{2}$. Thus, putting $z := \frac{x_0}{\|x_0\|} \in S_X$, we have that $\|z - x_0\| < \frac{\varepsilon}{2}$. Since the mapping $x \mapsto \text{dc}(x)$ is 1-Lipschitz (see Remark 2.2), we have

$$\text{dc}(z) \geq \text{dc}(x_0) - |\text{dc}(z) - \text{dc}(x_0)| \geq \text{dc}(x_0) - \|z - x_0\| > \left(2 - \frac{\varepsilon}{2}\right) - \frac{\varepsilon}{2} = 2 - \varepsilon;$$

which implies $\sup_{x \in S_X} \text{dc}(x) = 2$. $\square$

Let us point out that it is unclear whether or not an analog result for almost Δ -points holds in general. In fact, given a point $x \in B_X$ satisfying that $x \in \overline{\text{co}} \Delta_\varepsilon(x)$, we cannot expect the normalization $z = \|x\|^{-1}x$ of the point x to satisfy $z \in \overline{\text{co}} \Delta_\varepsilon(z)$. Moreover, the mapping $x \in B_X \mapsto \delta c(x)$ is not a Lipschitz map in general (see Remark 3.4 for instance). Nevertheless, we note for certain Banach spaces that a result similar to Proposition 2.12 holds.

Proposition 2.13. *Let X be a Banach space such that $X \stackrel{1}{=} X \oplus_\infty \mathbb{R}$. Then X admits almost Δ -points if and only if $\sup_{x \in S_X} \delta c(x) = 2$.*

Definitions Used

For $x \in B_X$, the Δ -constant $\delta c(x)$ is the largest $\alpha \geq 0$ such that for every $\varepsilon, \delta > 0$ and every $x^* \in S_{X^*}$ with $x \in S(x^*, \delta)$, there is $y \in S(x^*, \delta)$ satisfying

$$\|y - x\| > \alpha - \varepsilon.$$

Choi and Jung prove the following characterization, used below: for $0 < \alpha \leq 2$,

$$\delta c(x) \geq \alpha \iff x \in \overline{\text{co}} \Delta_{2-\alpha+\varepsilon}(x) \text{ for every } \varepsilon > 0.$$

They also prove that $\delta c(x) = 0$ exactly at denting points of B_X .

The Counterexample

Theorem 1. *There exists a Banach space X which admits almost Δ -points, but*

$$\sup_{x \in S_X} \delta c(x) = 0.$$

Consequently, the implication

$$X \text{ admits almost } \Delta\text{-points} \implies \sup_{x \in S_X} \delta c(x) = 2$$

is false in general.

Proof Intuition

Interior points of high-dimensional cubes can be almost Δ -points: the point $(1 - 2/m, \dots, 1 - 2/m)$ is the average of the m sign vectors with one negative coordinate, and each of those sign vectors is almost distance 2 from it. The unit balls of ℓ_p^m with p very large are almost cubes, so this convex-combination witness survives after a small scaling.

To make the unit sphere behave in the opposite way, put these blocks together with an ℓ_2 -sum. Since every finite ℓ_p^m , $1 < p < \infty$, is locally uniformly rotund and an ℓ_2 -sum of such blocks is locally uniformly rotund, every point of the resulting unit sphere is denting. Choi and Jung's denting-point characterization then forces $\delta c = 0$ on the whole unit sphere, while the interior block witnesses still give almost Δ -points.

Proof. For each integer $m \geq 3$, let

$$p_m = m^2, \quad E_m = \ell_{p_m}^m, \quad D_m = m^{1/p_m}.$$

Let

$$X = \left(\bigoplus_{m=3}^{\infty} E_m \right)_{\ell_2}.$$

First we show that X has almost Δ -points. In ℓ_{∞}^m , put

$$t_m = 1 - \frac{2}{m}, \quad u_m = (t_m, \dots, t_m).$$

For $1 \leq i \leq m$, let $w_i^{(m)} \in \{-1, 1\}^m$ be the sign vector whose i -th coordinate is -1 and whose other coordinates are 1. Then

$$\frac{1}{m} \sum_{i=1}^m w_i^{(m)} = u_m$$

and

$$\|w_i^{(m)} - u_m\|_{\infty} = 2 - \frac{2}{m} \quad (1 \leq i \leq m).$$

Regard these vectors in $E_m = \ell_{p_m}^m$ and scale by D_m . Since $\|w_i^{(m)}\|_{p_m} = D_m$, the vectors

$$z_i^{(m)} = \frac{w_i^{(m)}}{D_m}$$

belong to B_{E_m} . Also

$$x_m = \frac{u_m}{D_m} \in B_{E_m}, \quad x_m = \frac{1}{m} \sum_{i=1}^m z_i^{(m)}.$$

Moreover,

$$\|z_i^{(m)} - x_m\|_{p_m} = \frac{1}{D_m} \|w_i^{(m)} - u_m\|_{p_m} \geq \frac{1}{D_m} \|w_i^{(m)} - u_m\|_{\infty} = \frac{2 - 2/m}{m^{1/m^2}}.$$

By the convex-hull characterization of the Δ -constant,

$$\delta c_{E_m}(x_m) \geq \beta_m := \frac{2 - 2/m}{m^{1/m^2}}.$$

Since $\beta_m \rightarrow 2$, the block spaces contain points whose Δ -constants tend to 2.

Embed E_m into X as the m -th coordinate. The same convex combination, now viewed in B_X , preserves all distances among the involved points. Hence

$$\delta c_X(\hat{x}_m) \geq \beta_m \rightarrow 2,$$

where $\hat{x}_m$ denotes x_m placed in the m -th coordinate of the sum. Therefore $\sup_{x \in B_X} \delta c_X(x) = 2$, equivalently X admits almost Δ -points.

It remains to prove that every point of S_X has Δ -constant zero. Each $E_m = \ell_{p_m}^m$ is locally uniformly rotund. The ℓ_2 -sum X is also locally uniformly rotund. Indeed, suppose $a^k, a \in S_X$ and $\|a^k + a\| \rightarrow 2$. Let

$$r_m^k = \|a_m^k\|_{E_m}, \quad r_m = \|a_m\|_{E_m}.$$

Then

$$\|a^k + a\|_X \leq \|(r_m^k + r_m)_m\|_{\ell_2} \leq 2,$$

so $\|(r_m^k + r_m)_m\|_2 \rightarrow 2$. Since ℓ_2 is locally uniformly rotund, $(r_m^k)_m \rightarrow (r_m)_m$ in ℓ_2 . For each coordinate with $r_m > 0$, the normalized vectors in E_m satisfy the local-uniform-rotundity criterion in E_m , and hence $a_m^k \rightarrow a_m$. Combining this coordinatewise convergence on a large finite set with the ℓ_2 -smallness of the tails gives $\|a^k - a\|_X \rightarrow 0$. Thus X is locally uniformly rotund.

Every point of the unit sphere of a locally uniformly rotund space is a denting point of the unit ball: a norming slice at the point has arbitrarily small diameter by the defining local uniform rotundity implication. Choi and Jung's Proposition 2.9 identifies denting points of B_X with points $x \in S_X$ such that $\delta c(x) = 0$. Therefore

$$\delta c_X(x) = 0 \quad (x \in S_X),$$

and $\sup_{x \in S_X} \delta c_X(x) = 0$. □

Verification Notes

The construction is analytic. The included script `code/verify_constants.py` only prints sample values of

$$\beta_m = \frac{2 - 2/m}{m^{1/m^2}},$$

confirming numerically that the lower bounds tend upward to 2. It is not used as proof.

Novelty and Literature Check

The run indexes were searched for arXiv:2307.10647, Daugavet and Δ -constants, almost Δ -points, and related Delta-point phrases. Only adjacent packets about superreflexive Δ -points appeared; no packet covered this question.

An external web search for exact phrases around “almost Delta-points”, “Delta-constant”, and the source’s “unclear whether” passage found the source paper but no later answer. Novelty confidence is therefore moderate: the construction appears not to be recorded in the run memory or surfaced by a bounded search, but the literature check was not exhaustive.

Limitations

This does not challenge Choi and Jung's positive Proposition 2.13 for spaces isomorphic to $X \oplus_\infty \mathbb{R}$. The constructed space is a Hilbertian sum of finite-dimensional ℓ_p spaces with $p \rightarrow \infty$, and is locally uniformly rotund.

Human Review Recommendation

Reviewers should check two points first: the transfer of the cube convex combination from ℓ_∞^m to $\ell_{p_m}^m$ after scaling by m^{1/p_m} , and the proof that the ℓ_2 -sum of the blocks is locally uniformly rotund despite the lack of a uniform modulus over m .

References

- [1] G. Choi and M. Jung, *The Daugavet and Delta-constants of points in Banach spaces*, arXiv:2307.10647.